



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded .

ISE-I Examination, Academic Year 2025-26 (Sem1)

Class: Second Year (Electronics and Telecommunication Engineering)

Date: 19/08/25

Time: 11 to 11:45 p.m.

Subject: Digital Electronics and Microprocessors
Subject code: PCC-EC202

Max. Marks : 20
Duration: 45 minutes

Note: I. Use of calculator is not allowed II Solve any four questions

Sr.No.	Questions	Marks
Q.1	What is Excess 3 and gray code. State applications of each code.	(5)
Q.2	Draw truth table of full adder and design using gates.	(5)
Q.3	Reduce the following function using K- map and implement using NOR gates. $F1 = \sum m(0,1,2,3, 4,5,6, 8,10,15)$	(5)
Q.4	State laws of Boolean Algebra. Prove Demorgan's laws.	(5)
Q.5	Binary number (001001001.101) is -----in hexadecimal and ----- in octal number. Hexadecimal number (FF) is ----- in decimal. Addition of octal numbers 15 and 12 is-----	(5)

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